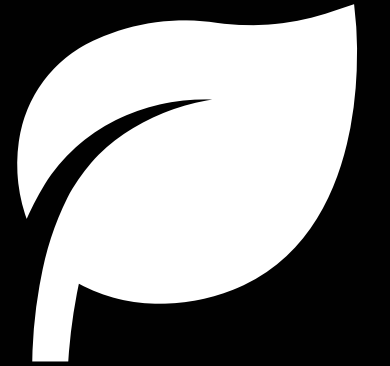


Generative AI

Session 04



Presented by Vansh Sood





From hackathons to the Startup Residency program, ShiftKey Labs makes sure students have the skills & support they need to pursue their most innovative ideas!



HIGHLIGHTS

NASA Space Apps Challenge

Global Game Jam

Generative AI Hackathon

Workshops & Talks

REACT
Blockchain
Prototyping
Startups
Freelancing
Cybersecurity

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About Me

- Third-year BCS, Head TA at Dalhousie University
- Tech coordinator for Shiftkey Labs and Faculty of Health
- Built Gen AI startup in Australia
- LinkedIn Top Web Development Voice
- Freelancing for over 7 years in most CS disciplines
- Worked with 25+ startups and small businesses
- Helped startups raise over \$100,000 in funding
- Founder of a web and mobile development agency
- AWS Certified Cloud Practitioner
- Google Code-In Grand Prize Winner (Global Top 50)



Vansh Sood

CS at Dalhousie University | Head TA | Tech
Founder | AWS Certified | Google Code In G...



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Course Overview

This will be an introductory course to Generative AI and Large Language Models. After successfully completing this certification, you'll have good knowledge of the following topics:

- Large Language Models
- History and working of these models
- Prompt Engineering
- Fine Tuning
- Hugging Face
- AI Ethics

What are you taking away from this?

Apart from the certificate? A whole lot of knowledge.

You can choose to just listen, or you can choose to actually work on AI tools to learn techniques to help you succeed

Work during these four weeks, and I guarantee you will have an amazing project with a whole lot of skills up your sleeve.

And even though I'm not a big fan of resumes, this will be an excellent addition to your resume.

Generative AI

Generative AI refers to a type of artificial intelligence that creates new content, such as text, images, audio, or video, by learning patterns from existing data. Instead of just analyzing or classifying data, generative AI models generate new examples that resemble the training data.

A generative AI model trained on thousands of images of landscapes can create entirely new, realistic-looking landscape images that have never existed before.



Large Language Model

Large Language Models (LLMs) are a type of AI model that is trained on vast amounts of text data to understand and generate human-like language.

These models use neural networks, to process and produce text based on the patterns they have learned from the data.

Applications: Used in chatbots, content generation, translation services, and more.

Capabilities: Can perform tasks like summarization, text completion, and generating creative content, often with little to no task-specific training.



Fine-tuning

Fine-tuning is the process of taking a pre-trained AI model and adapting it to a specific task or domain by training it on new, task-specific data.

Fine-tuning makes general models more suitable for niche tasks.

Example: A language model trained broadly on general text can be fine-tuned to better understand medical terminology or legal documents by training it with data from those domains.

Drawbacks of Fine-tuning

- Time and Resources: Requires additional data, computing power, and training time.
- Overfitting: Risk of becoming too specialized, reducing the model's ability to generalize.
- Maintenance: Needs frequent updates and retraining as new data becomes available.
- Knowledge Loss: May weaken general-purpose knowledge after being fine-tuned on specific tasks.

Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a hybrid approach in NLP that combines the power of information retrieval with text generation models.

It retrieves relevant documents from a knowledge base to provide additional context, and then uses a language model to generate more accurate and informed responses based on the retrieved information.

How it works

- User Input: The user provides a query or question.
- Retrieval: The system searches a knowledge base for relevant information.
- Augmentation: The retrieved information is fed into the language model.
- Generation: The model generates a response, using both the query and the retrieved data.
- Output: The user receives a factually grounded, informed response.

Fine Tuning vs RAG

Fine-tuning

Alters the model permanently

Requires a specific dataset

Resource-heavy and time-consuming

Requires retraining for updates

RAG

Does not change the model

Uses external data on demand

More efficient by retrieving data dynamically

Pulls updated data without retraining

Ethical Considerations

Microsoft lays off an ethical AI team as it doubles down on OpenAI

Rebecca Bellan @rebeccabellan / 12:26 PM GMT+11 • March 14, 2023

[Comment](#)



WHERE WE ARE GOING



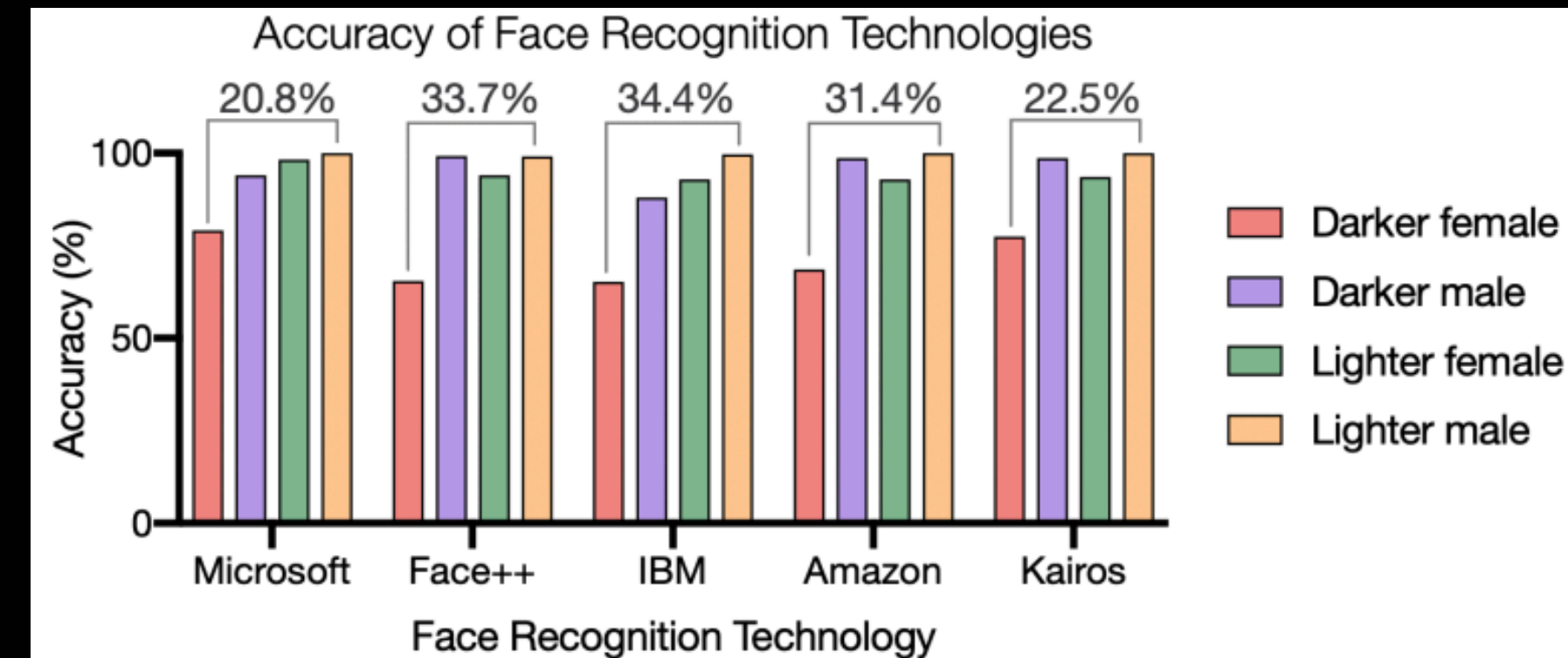
WE DONT NEED ETHICS

Bias

Bias in AI occurs when models produce unfair outcomes because they reflect biased patterns present in their training data.

An AI used in hiring might favor male candidates if its training data mostly consists of successful male applicants, leading to unfair gender discrimination.

<https://sitn.hms.harvard.edu/flash/2020/racial-discrimination-in-face-recognition-technology/>



Google reportedly targeted people with 'dark skin' to improve facial recognition

Subcontracted workers were told to persuade subjects to agree to face scans, mischaracterizing them as a 'selfie game' or 'survey'

Accountability

Accountability in AI refers to determining who is responsible when AI systems cause harm or make mistakes.

If an autonomous vehicle causes an accident, it's unclear whether the car manufacturer, the software developer, or the owner is accountable.

It's like a self-driving bus causing an accident—who takes responsibility: the driver, the bus manufacturer, or the software developers? Similarly, AI accountability is a challenge that needs clear frameworks.

Cruise recalls all self-driving cars after grisly accident and California ban

All 950 of the General Motors subsidiary's autonomous cars will be taken off roads for a software update

"In the crash, another vehicle with a person behind the wheel struck a pedestrian, sending the person into the path of a Cruise autonomous vehicle. The Cruise initially stopped but still hit the person. But it then pulled to the right to get out of traffic, pulling the person about 20ft (six meters) forward. The pedestrian was pinned under one of the Cruise vehicle's tires and was critically injured."

Security Concerns

AI systems can be vulnerable to hacking, manipulation, or misuse, leading to security risks like data breaches or malicious behavior.

AI-powered facial recognition systems could be hacked to gain unauthorized access to secure areas or sensitive information.

Me after using ChatGPT to pass the bar exam and getting my first client sentenced to death



Mitigating Skeleton Key, a new type of generative AI jailbreak technique

This AI jailbreak technique works by using a multi-turn (or multiple step) strategy to cause a model to ignore its guardrails. Once guardrails are ignored, a model will not be able to determine malicious or unsanctioned requests from any other. Because of its full bypass abilities, we have named this jailbreak technique Skeleton Key.

Jobs

pretty self explanatory

Responsible AI by Microsoft (haha)

- **Fairness:** AI systems should treat everyone equally and provide the same recommendations to all individuals. Fairness in AI systems prevents discrimination based on personal characteristics.
- **Reliability and safety:** AI systems must operate reliably, safely, and consistently under various conditions to help build trust.
- **Privacy and security:** AI systems should respect privacy and maintain security by protecting private and confidential information. They should also resist attacks and attempts to corrupt or compromise the system.
- **Inclusiveness:** AI systems should empower and engage everyone. Inclusive design practices can help AI system developers understand and address potential exclusion barriers in a product or service. Inclusiveness fosters innovation and helps design experiences that benefit everyone.
- **Transparency:** AI systems should be transparent and understandable. AI systems can inform decisions that can deeply affect people's lives, so it's crucial for individuals to understand how the system makes these decisions.
- **Accountability:** AI systems and their developers should be accountable and answerable.

EU AI Act

- **Prohibited Practices:** The Act explicitly prohibits AI systems that manipulate human behavior in harmful ways or employ deceptive techniques.
- **High-Risk Classification:** High-risk AI systems must undergo conformity assessments before being placed on the market. Providers must implement robust risk management and transparency measures.
- **Governance Structure:** The European AI Office will oversee enforcement and implementation, ensuring compliance across member states and fostering international cooperation on AI governance



More than 100 companies signed EU Pact for AI. Meta and Apple missing from the list

The signatories pledge to implement the European Artificial Intelligence Act requirements in advance. Two key pillars of the pact proposed by the Commission: collaboration and compliance with the AI regulations



by **Noemi Morucci** — 25 September 2024 in Digital, In the spotlight

Thank you

Reach out to me:

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